

A Choose correct answer from the given options. [Each carries 1 Mark] [27]

- Which of the following substance has highest solubility in water ?
(A) Ethanol (B) Ethylene glycol (C) Glycerine (D) Propanol
- On increasing temperature, the value of Henry's constant
(A) decreases (B) increases (C) remains constant (D) uncertain
- 0.5 M glucose solution is iso-osmotic with which of the following solutions ?
(A) 0.10 M NaCl (B) 0.05 NaCl (C) 0.25 M NaCl (D) 1 M NaCl
- The real vapour pressure for non ideal solution is than by Raoult's law.
(A) more (B) less (C) equal (D) (A) and (B) both
- The total pressure for volatile components A and B is 0.02 bar at equilibrium. If the mole fractions at component A is 0.2, then, what will be the partial pressure of component B ?
(A) 0.02 bar (B) 0.04 bar (C) 0.016 bar (D) 0.2 bar
- Soda water is of which type solution ?
(A) Solid solution (B) Liquid solution (C) Gaseous solution (D) None of these
- Which of the following is non aqueous solvent ?
(A) Benzene (B) Ether (C) CCl_4 (D) All of these
- Pickup the correct formula of potash alum.
(A) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24 \text{H}_2\text{O}$ (B) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_2 \cdot 24 \text{H}_2\text{O}$
(C) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 12 \text{H}_2\text{O}$ (D) $\text{K}_2\text{SO}_4 \cdot \text{Al}(\text{SO}_4)_3 \cdot 12 \text{H}_2\text{O}$
- Formality unit is used in which scientific field ?
(A) Pharmacy (B) Medical (C) Agriculture (D) (A) and (B) both
- What is the nature of graph of mole fraction of gaseous solute and its partial pressure ?
(A) Straight line (B) Parabola (C) Zigzag line (D) None of these
- 1 ppm = mass of solute in 1 liter solution.
(A) 10^{-3} gram (B) 10^{-4} gram (C) 10^{-6} gram (D) 10^{-2} gram
- What is the use of Henry's law ?
(A) In preparation of cold drinks (B) In biochemical processes
(C) In the industry (D) All of these
- Which of the following statement is wrong ?
(A) K_H increase with increase in temperature.
(B) With increase in temperature, the solubility of gaseous solute in liquid decrease.
(C) K_H remains constant with change in temperature.
(D) $p \propto X$
- Which are the affecting factors on the vapour pressure ?
(A) The nature of liquid (B) Temperature (C) (A) and (B) both (D) None of these
- According to Raoult's law, for which system of solution, the vapour pressure of both i.e. solute and solvent will be proportional to their mole fraction ?
(A) Both are volatile (B) Solute is volatile, solvent is non volatile
(C) Both are nonvolatile (D) Solute is nonvolatile, solvent is volatile
- Which of the following pair is of non ideal solution ?
(A) $\text{HCl} + \text{H}_2\text{O}$ (B) $\text{HNO}_3 + \text{H}_2\text{O}$ (C) $\text{C}_6\text{H}_5\text{OH} + \text{C}_6\text{H}_5\text{NH}_2$ (D) All of these
- The solution which do not obey the Raoult's law over the entire range of concentration is called

which solution ?

- (A) Ideal (B) Non Ideal (C) True Solution (D) (B) and (C) both

18. Which is not a natural semi permeable membrane ?

- (A) Plant cellwall (B) Urinary bladder (C) Cellophane paper (D) None of these

19. Which relation is true, if molecular weight of solute is x and its weight is y ?

- (A) $x = \left(\frac{y}{v}\right) \frac{RT}{\pi}$ (B) $y = \left(\frac{x}{y}\right) \frac{RT}{\pi}$ (C) $x = \left(\frac{v}{y}\right) \frac{RT}{\pi}$ (D) $x = \left(\frac{y}{v}\right) \frac{R\pi}{T}$

20. At 273 K temperature, if 10 gm glucose (p_1), 10 gm urea (p_2) and 10 gm sucrose (p_3) is dissolved in 250 mL of water, then what is the correct relation of osmotic pressure for them ?

- (A) $p_1 > p_2 > p_3$ (B) $p_3 > p_2 > p_1$ (C) $p_2 > p_1 > p_3$ (D) $p_2 > p_3 > p_1$

21. If osmotic pressure of an aqueous solution of sucrose having concentration of 34.2 gm/liter at 17°C is 2.38 atmosphere, then gm/L glucose is isotonic with this solution.

- (A) 34.2 (B) 17.1 (C) 36.0 (D) 18.0

22. To produce difference between freezing point and boiling point of a solution by 105.0°C, how much sucrose should be dissolved in 100 gm of water ? ($K_f = 1.86^\circ\text{C} \cdot \text{kg mol}^{-1}$ and $K_b = 0.51^\circ\text{C kg mol}^{-1}$)

- (A) 72 gm (B) 34.2 gm (C) 342 gm (D) 460 gm

23. 20% aqueous FeCl_3 solution has density of 1.1 gm/mL then molar concentration of this solution is.....

- (A) 0.028 (B) 1.357 (C) 0.163 (D) 1.47

24. To prepare solution of certain substance, 300 gm is used for 25% solution and 400 gm for 40% solution, then what is the mass percentage of solute present in this solution mixture ?

- (A) 33.57 (B) 66.43 (C) 87.23 (D) 19.24

25. Give mole fraction of solute present in 1.00 m aqueous solution.

- (A) 0.177 (B) 0.00177 (C) 0.0177 (D) 0.034

26. Find out molarity of NO_3^- ions in solution when mixing 2 litre 3 M AgNO_3 solution with 3 litre 1 M BaCl_2 solution.

- (A) 1.2 M (B) 0.1 M (C) 0.5 M (D) 0.4 M

27. Which equimolal concentrated solution has highest freezing point ?

- (A) Glucose (B) $\text{La}(\text{NO}_3)_3$ (C) $\text{Ca}(\text{NO}_3)_2$ (D) Vinegar

BSA

NCERT level practice test

Std 12 Chemistry
chapters 6

Date : 26/05/26

Total Marks : 30

Time :

Section [A] : 1 Mark MCQ

No	Ans	Chap	Sec	Que	Universal_Queld
1.	C	Chap 1	S8	1	QP25P11B1212_P1C1S8Q1
2.	B	Chap 1	S8	2	QP25P11B1212_P1C1S8Q2
3.	C	Chap 1	S8	3	QP25P11B1212_P1C1S8Q3
4.	D	Chap 1	S8	4	QP25P11B1212_P1C1S8Q4
5.	C	Chap 1	S8	5	QP25P11B1212_P1C1S8Q5
6.	B	Chap 1	S8	6	QP25P11B1212_P1C1S8Q6
7.	D	Chap 1	S8	7	QP25P11B1212_P1C1S8Q7
8.	A	Chap 1	S8	8	QP25P11B1212_P1C1S8Q8
9.	D	Chap 1	S8	9	QP25P11B1212_P1C1S8Q9
10.	A	Chap 1	S8	10	QP25P11B1212_P1C1S8Q10
11.	A	Chap 1	S8	11	QP25P11B1212_P1C1S8Q11
12.	D	Chap 1	S8	12	QP25P11B1212_P1C1S8Q12
13.	C	Chap 1	S8	13	QP25P11B1212_P1C1S8Q13
14.	C	Chap 1	S8	14	QP25P11B1212_P1C1S8Q14
15.	A	Chap 1	S8	15	QP25P11B1212_P1C1S8Q15
16.	D	Chap 1	S8	16	QP25P11B1212_P1C1S8Q16
17.	D	Chap 1	S8	17	QP25P11B1212_P1C1S8Q17
18.	C	Chap 1	S8	18	QP25P11B1212_P1C1S8Q18
19.	A	Chap 1	S8	19	QP25P11B1212_P1C1S8Q19
20.	C	Chap 1	S8	21	QP25P11B1212_P1C1S8Q21
21.	D	Chap 1	S8	22	QP25P11B1212_P1C1S8Q22
22.	A	Chap 1	S8	20	QP25P11B1212_P1C1S8Q20
23.	B	Chap 1	S8	24	QP25P11B1212_P1C1S8Q24
24.	A	Chap 1	S8	23	QP25P11B1212_P1C1S8Q23
25.	C	Chap 1	S8	25	QP25P11B1212_P1C1S8Q25
26.	A	Chap 1	S8	26	QP25P11B1212_P1C1S8Q26
27.	A	Chap 1	S8	27	QP25P11B1212_P1C1S8Q27

BSA				
NCERT level practice test		Std 12 Chemistry		Date : 26/05/26
Total Marks : 27		chapters 6		Time : 01 Hour
Student Name : _____				
Roll Number : 012 0123456789 0123456789				
A B C D 1. ☐ ☐ ☐ ☐ 2. ☐ ☐ ☐ ☐ 3. ☐ ☐ ☐ ☐ 4. ☐ ☐ ☐ ☐ 5. ☐ ☐ ☐ ☐ 6. ☐ ☐ ☐ ☐ 7. ☐ ☐ ☐ ☐ 8. ☐ ☐ ☐ ☐ 9. ☐ ☐ ☐ ☐ 10. ☐ ☐ ☐ ☐ <td> A B C D 11. ☐ ☐ ☐ ☐ 12. ☐ ☐ ☐ ☐ 13. ☐ ☐ ☐ ☐ 14. ☐ ☐ ☐ ☐ 15. ☐ ☐ ☐ ☐ 16. ☐ ☐ ☐ ☐ 17. ☐ ☐ ☐ ☐ 18. ☐ ☐ ☐ ☐ 19. ☐ ☐ ☐ ☐ 20. ☐ ☐ ☐ ☐ <td> A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ <td></td><td></td></td></td>	A B C D 11. ☐ ☐ ☐ ☐ 12. ☐ ☐ ☐ ☐ 13. ☐ ☐ ☐ ☐ 14. ☐ ☐ ☐ ☐ 15. ☐ ☐ ☐ ☐ 16. ☐ ☐ ☐ ☐ 17. ☐ ☐ ☐ ☐ 18. ☐ ☐ ☐ ☐ 19. ☐ ☐ ☐ ☐ 20. ☐ ☐ ☐ ☐ <td> A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ <td></td><td></td></td>	A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ <td></td> <td></td>		

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BSA				
NCERT level practice test		Std 12 Chemistry		Date : 26/05/26
Total Marks : 27		chapters 6		Time : 01 Hour
Student Name : _____				
Roll Number : 012 0123456789 0123456789				
A B C D 1. ☐ ☐ ☐ ☐ 2. ☐ ☐ ☐ ☐ 3. ☐ ☐ ☐ ☐ 4. ☐ ☐ ☐ ☐ 5. ☐ ☐ ☐ ☐ 6. ☐ ☐ ☐ ☐ 7. ☐ ☐ ☐ ☐ 8. ☐ ☐ ☐ ☐ 9. ☐ ☐ ☐ ☐ 10. ☐ ☐ ☐ ☐ <td> A B C D 11. ☐ ☐ ☐ ☐ 12. ☐ ☐ ☐ ☐ 13. ☐ ☐ ☐ ☐ 14. ☐ ☐ ☐ ☐ 15. ☐ ☐ ☐ ☐ 16. ☐ ☐ ☐ ☐ 17. ☐ ☐ ☐ ☐ 18. ☐ ☐ ☐ ☐ 19. ☐ ☐ ☐ ☐ 20. ☐ ☐ ☐ ☐ <td> A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ <td></td><td></td></td></td>	A B C D 11. ☐ ☐ ☐ ☐ 12. ☐ ☐ ☐ ☐ 13. ☐ ☐ ☐ ☐ 14. ☐ ☐ ☐ ☐ 15. ☐ ☐ ☐ ☐ 16. ☐ ☐ ☐ ☐ 17. ☐ ☐ ☐ ☐ 18. ☐ ☐ ☐ ☐ 19. ☐ ☐ ☐ ☐ 20. ☐ ☐ ☐ ☐ <td> A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ <td></td><td></td></td>	A B C D 21. ☐ ☐ ☐ ☐ 22. ☐ ☐ ☐ ☐ 23. ☐ ☐ ☐ ☐ 24. ☐ ☐ ☐ ☐ 25. ☐ ☐ ☐ ☐ 26. ☐ ☐ ☐ ☐ 27. ☐ ☐ ☐ ☐ <td></td> <td></td>		

A Choose correct answer from the given options. [Each carries 1 Mark] [27]

1. Which of the following substance has highest solubility in water ?
 (A) Ethanol (B) Ethylene glycol (C) Glycerine (D) Propanol

Ans. (C) Glycerine

→ $\begin{array}{c} \text{CH}_2 - \text{CH} - \text{CH}_2 \\ | \quad | \quad | \\ \text{OH} \quad \text{OH} \quad \text{OH} \end{array}$ As glycerine having more -OH group, it is highly soluble in water.

2. On increasing temperature, the value of Henry's constant
 (A) decreases (B) increases (C) remains constant (D) uncertain

Ans. (B) increases

→ With increase in temp (T), K_H will also increase and solubility of gaseous solute decreases.

3. 0.5 M glucose solution is iso-osmotic with which of the following solutions ?
 (A) 0.10 M NaCl (B) 0.05 NaCl (C) 0.25 M NaCl (D) 1 M NaCl

Ans. (C) 0.25 M NaCl

4. The real vapour pressure for non ideal solution is than by Raoult's law.
 (A) more (B) less (C) equal (D) (A) and (B) both

Ans. (D) (A) and (B) both

5. The total pressure for volatile components A and B is 0.02 bar at equilibrium. If the mole fractions at component A is 0.2, then, what will be the partial pressure of component B ?
 (A) 0.02 bar (B) 0.04 bar (C) 0.016 bar (D) 0.2 bar

Ans. (C) 0.016 bar

→ $X_A + X_B = 1$

$$\therefore 0.2 + X_B = 1 \quad \therefore X_B = 0.8$$

$$p_B = p \cdot X_B = (0.02) (0.8)$$

$$\therefore p_B = 0.016 \text{ bar}$$

6. Soda water is of which type solution ?
 (A) Solid solution (B) Liquid solution (C) Gaseous solution (D) None of these

Ans. (B) Liquid solution

7. Which of the following is non aqueous solvent ?
 (A) Benzene (B) Ether (C) CCl_4 (D) All of these

Ans. (D) All of these

8. Pickup the correct formula of potash alum.
 (A) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24 \text{H}_2\text{O}$ (B) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_2 \cdot 24 \text{H}_2\text{O}$
 (C) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 12 \text{H}_2\text{O}$ (D) $\text{K}_2\text{SO}_4 \cdot \text{Al}(\text{SO}_4)_3 \cdot 12 \text{H}_2\text{O}$

Ans. (A) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24 \text{H}_2\text{O}$

9. Formality unit is used in which scientific field ?
 (A) Pharmacy (B) Medical (C) Agriculture (D) (A) and (B) both

Ans. (D) (A) and (B) both

10. What is the nature of graph of mole fraction of gaseous solute and its partial pressure ?
 (A) Straight line (B) Parabola (C) Zigzag line (D) None of these

Ans. (A) Straight line

11. 1 ppm = mass of solute in 1 liter solution.

- (A) 10^{-3} gram (B) 10^{-4} gram (C) 10^{-6} gram (D) 10^{-2} gram

Ans. (A) 10^{-3} gram

12. What is the use of Henry's law ?

- (A) In preparation of cold drinks (B) In biochemical processes
(C) In the industry (D) All of these

Ans. (D) All of these

13. Which of the following statement is wrong ?

- (A) K_H increase with increase in temperature.
(B) With increase in temperature, the solubility of gaseous solute in liquid decrease.
(C) K_H remains constant with change in temperature.
(D) $p \propto X$

Ans. (C) K_H remains constant with change in temperature.

14. Which are the affecting factors on the vapour pressure ?

- (A) The nature of liquid (B) Temperature (C) (A) and (B) both (D) None of these

Ans. (C) (A) and (B) both

15. According to Raoult's law, for which system of solution, the vapour pressure of both i.e. solute and solvent will be proportional to their mole fraction ?

- (A) Both are volatile (B) Solute is volatile, solvent is non volatile
(C) Both are nonvolatile (D) Solute is nonvolatile, solvent is volatile

Ans. (A) Both are volatile

16. Which of the following pair is of non ideal solution ?

- (A) $\text{HCl} + \text{H}_2\text{O}$ (B) $\text{HNO}_3 + \text{H}_2\text{O}$ (C) $\text{C}_6\text{H}_5\text{OH} + \text{C}_6\text{H}_5\text{NH}_2$ (D) All of these

Ans. (D) All of these

17. The solution which do not obey the Raoult's law over the entire range of concentration is called which solution ?

- (A) Ideal (B) Non Ideal (C) True Solution (D) (B) and (C) both

Ans. (D) (B) and (C) both

18. Which is not a natural semi permeable membrane ?

- (A) Plant cellwall (B) Urinary bladder (C) Cellophane paper (D) None of these

Ans. (C) Cellophane paper

19. Which relation is true, if molecular weight of solute is x and its weight is y ?

- (A) $x = \left(\frac{y}{v}\right) \frac{RT}{\pi}$ (B) $y = \left(\frac{x}{y}\right) \frac{RT}{\pi}$ (C) $x = \left(\frac{v}{y}\right) \frac{RT}{\pi}$ (D) $x = \left(\frac{y}{v}\right) \frac{R\pi}{T}$

Ans. (A) $x = \left(\frac{y}{v}\right) \frac{RT}{\pi}$

20. At 273 K temperature, if 10 gm glucose (p_1), 10 gm urea (p_2) and 10 gm sucrose (p_3) is dissolved in 250 mL of water, then what is the correct relation of osmotic pressure for them ?

- (A) $p_1 > p_2 > p_3$ (B) $p_3 > p_2 > p_1$ (C) $p_2 > p_1 > p_3$ (D) $p_2 > p_3 > p_1$

Ans. (C) $p_2 > p_1 > p_3$

➡ $p = \frac{wRT}{MV}$ Where, $\frac{wRT}{V}$ is constant and so, $p \propto \frac{1}{M}$

So, $p_2 > p_1 > p_3$

21. If osmotic pressure of an aqueous solution of sucrose having concentration of 34.2 gm/liter at 17°C is 2.38 atmosphere, then gm/L glucose is isotonic with this solution.

- (A) 34.2 (B) 17.1 (C) 36.0 (D) 18.0

Ans. (D) 18.0

➡ For isotonic solutions, concentration should be same, so concentration of sucrose solution at 34.2 gm/liter is 0.1 M.

➡ So concentration 18.0 gm/liter glucose is 0.1 M.

22. To produce difference between freezing point and boiling point of a solution by 105.0°C , how much sucrose should be dissolved in 100 gm of water ? ($K_f = 1.86^{\circ}\text{C} \cdot \text{kg mol}^{-1}$ and $K_b = 0.51^{\circ}\text{C kg mol}^{-1}$)
- (A) 72 gm (B) 34.2 gm (C) 342 gm (D) 460 gm

Ans. (A) 72 gm

➡ Boiling point (T_b) = $100 + \Delta T_b$
= $100 + K_b \cdot m$

➡ Freezing point (T_f) = $0 - \Delta T_f = -K_f \cdot m$

$$\therefore T_b - T_f = (100 + K_b \cdot m) - (-K_f \cdot m)$$

$$105 = 100 + 0.51 m + 1.86 m$$

$$2.37 m = 5$$

$$m = \frac{5}{2.37} = 2.11$$

➡ So, mass of sucrose to dissolve in 100 gm water is, = $\frac{2.11 \times 342 \times 100}{1000} = 72 \text{ gm}$

23. 20% aqueous FeCl_3 solution has density of 1.1 gm/mL then molar concentration of this solution is.....
- (A) 0.028 (B) 1.357 (C) 0.163 (D) 1.47

Ans. (B) 1.357

➡ 20% FeCl_3 solution means 20 gm FeCl_3 is present in 100 gm of solution.

$$\therefore \text{Volume of 100 gm solution} = \frac{100}{1.1} = 90.91 \text{ mL}$$

$$\text{Moles of 20 gm } \text{FeCl}_3 \text{ solution} = \frac{20}{162} = 0.1234 \text{ mol}$$

$$\text{So, molar concentration of solution} = \frac{0.1234 \times 1000}{90.91} = 1.357 \text{ M}$$

24. To prepare solution of certain substance, 300 gm is used for 25% solution and 400 gm for 40% solution, then what is the mass percentage of solute present in this solution mixture ?
- (A) 33.57 (B) 66.43 (C) 87.23 (D) 19.24

Ans. (A) 33.57

➡ In solution-1, mass of solute = $\frac{25}{100} \times 300 = 75 \text{ gm}$

$$\text{In solution-2, mass of solute} = \frac{40}{100} \times 400 = 160 \text{ gm}$$

➡ So, total mass of solute on mixing two solution is = 235 gm

So, total mass of solution = 700 gm

$$\text{So, \% solute present in final solution} = \frac{235}{700} \times 100 = 33.57\%$$

25. Give mole fraction of solute present in 1.00 m aqueous solution.
- (A) 0.177 (B) 0.00177 (C) 0.0177 (D) 0.034

Ans. (C) 0.0177

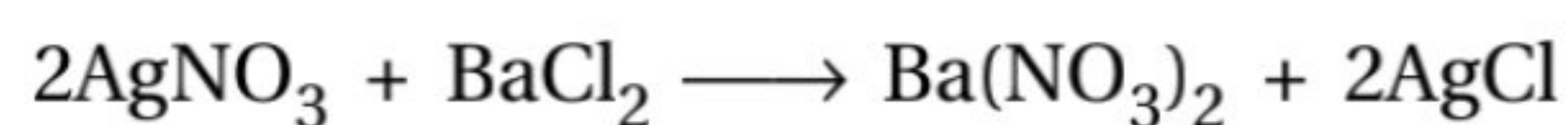
26. Find out molarity of NO_3^- ions in solution when mixing 2 litre 3 M AgNO_3 solution with 3 litre 1 M BaCl_2 solution.

- (A) 1.2 M (B) 0.1 M (C) 0.5 M (D) 0.4 M

Ans. (A) 1.2 M

⇒ 2 litre of 3M AgNO_3 solution has 3 moles of AgNO_3 .

3 litre of 1M BaCl_2 solution has 3 moles of BaCl_2 .



⇒ From above reaction, it is clear that 6 moles of AgNO_3 is reacted with 3 moles of BaCl_2 and produce 3 moles of $\text{Ba}(\text{NO}_3)_2$.

⇒ So in 5 litre solution total 6 moles of NO_3^- ions is produced.

$$\therefore \text{Molarity of } \text{NO}_3^- \text{ ions} = \frac{6}{5} = 1.2$$

27. Which equimolal concentrated solution has highest freezing point ?

- (A) Glucose (B) $\text{La}(\text{NO}_3)_3$ (C) $\text{Ca}(\text{NO}_3)_2$ (D) Vinegar

Ans. (A) Glucose